

Tarek Mahmoud

PRINCIPAL DATA SCIENTIST · NLP · LLMs · AGENTIC AI · PRESIGHT AT G42

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Summary

Principal Data Scientist and co-founder at Presight, G42, specializing in agentic AI, generative AI, retrieval-augmented generation, production machine learning, and big data analytics. Works across government, finance, and healthcare in professional services, product development, and R&D capacities, translating complex requirements into deployed AI systems and reusable technical capabilities. Holds a PhD from MBZUAI, with interdisciplinary research focused on modeling persuasion in language spanning psycholinguistics, cognitive computing, and computational social science.

Professional Experience

INDUSTRY

2023–present **Principal Data Scientist**, Presight (G42 Group), Abu Dhabi, UAE

2021–2023 **Senior Data Scientist**, Presight (G42 Group), Abu Dhabi, UAE

2019–2021 **Co-Founding Data Scientist**, Presight (G42 Group), Abu Dhabi, UAE

- Leading client-facing AI initiatives in fast-paced, agile environments across government, finance, public health, and behavioral analytics, serving as a product owner and translating stakeholder requirements into production systems across professional services, product development, and applied R&D.
- Developing core AI capabilities for Presight Lumina, Presight Vitruvian, and Presight Report Optimizer across law-enforcement, enterprise agentic AI, investment intelligence, and automated analytical reporting.
- Developing an Azure-based, real-time multi-speaker speech translation system combining ASR, machine translation, TTS, and diarization for live multilingual events and meeting intelligence.
- Advising senior leadership on AI strategy, technical planning, vendor selection, due diligence, and acquisition decisions, while leading and contributing to cross-functional technical committees.
- Selected by leadership to mentor VVIP participants as well as interns through data-science placements, designing applied projects and providing hands-on technical guidance.
- Built, deployed, and maintained production NLP, computer-vision, knowledge-graph, and cloud-based big-data pipelines processing heterogeneous, large-scale structured, and unstructured data.
- Spearheaded the data-science capabilities supporting Presight's national COVID-19 response, including forecasting, simulation, contact tracing, risk profiling, and vaccine prioritization.

2018 **Machine Learning Engineer (Internship)**, Aramco, Dhahran, KSA

RESEARCH

2022–2026 **Graduate Research Assistant**, MBZUAI, Abu Dhabi, UAE

- Published 15+ papers at ACL, EACL, NAACL, and EMNLP venues; 700+ citations, h-index 9.
- Co-organized three SemEval/GenAIDetect shared tasks on machine-generated text detection and narrative extraction.

2016–2019 **Graduate Research Assistant**, KAUST, Thuwal, KSA

2015 **Research Intern**, KAUST (Visual Computing Center), Thuwal, KSA

2012–2016 **Undergraduate Research Assistant**, KFUPM, Dhahran, KSA

TEACHING

- 2023–2025 **VVIP Mentor**, G42 Special Programs, Abu Dhabi, UAE — Selected by G42 & Presight leadership to deliver bespoke education and mentorship to high-profile VVIP participants, including members of prominent families and senior public figures. Designed and delivered tailored theoretical and practical sessions on generative AI, big data analytics, and cloud computing.
- 2025 **Teaching Assistant**, MBZUAI Research Internship Program, Abu Dhabi, UAE — mentored summer research interns, resulting in the FRaN-X demo paper
- 2024 **Teaching Assistant**, MBZUAI Research Internship Program, Abu Dhabi, UAE — mentored summer research interns, resulting in the EACL 2024 FRAPPE demo paper
- 2023–2025 **Teaching Assistant**, MBZUAI (Graduate NLP Course), Abu Dhabi, UAE
- 2018 **Teaching Assistant**, KAUST (Graduate ML Course), Thuwal, KSA

Education

Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI)

Abu Dhabi, UAE

PHD IN COMPUTER SCIENCE, NATURAL LANGUAGE PROCESSING

2022 – 2026

- Dissertation: Modeling Persuasion in Language
- Research Focus: Psycholinguistics, cognitive computing, computational social science

King Abdullah University of Science and Technology (KAUST)

Thuwal, KSA

MSC ELECTRICAL & COMPUTER ENGINEERING, AI SYSTEMS TRACK

2016 – 2017

King Fahd University of Petroleum & Minerals (KFUPM)

Dhahran, KSA

BSC ELECTRICAL ENGINEERING, FIRST HONORS, GPA 4.00/4.00

2012 – 2016

Publications

1. **Tarek Mahmoud**, Veronika Solopova, ..., and Preslav Nakov. “Uncovering Temporal Framing in the News”. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics: ACL 2026*. (Top 15%, Oral Presentation, SAC Highlights Award)
2. Yuxia Wang, Ruoyu Xing, ..., and **Tarek Mahmoud**, ..., and Preslav Nakov. “Is Human-Like Text Liked by Humans? Multilingual Human Detection and Preference Against AI”. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics: ACL 2026*. (Oral Presentation)
3. **Tarek Mahmoud**. “Modeling Persuasion in Language”. *PhD Dissertation, MBZUAI, 2026*.
4. Jakub Piskorski, **Tarek Mahmoud**, Nikolaos Nikolaidis, Ricardo Campos, Alípio Mário Jorge, Dimitar Dimitrov, ..., and Preslav Nakov. “SemEval 2025 Task 10: Multilingual Characterization and Extraction of Narratives from Online News”. In *Proceedings of the 19th International Workshop on Semantic Evaluation: SemEval 2025*.
5. **Tarek Mahmoud**, Zhuohan Xie, Dimitar Iliyanov Dimitrov, Nikolaos Nikolaidis, Purificação Silvano, Roman Yangarber, ..., and Preslav Nakov. “Entity Framing and Role Portrayal in the News”. In *Findings of the Association for Computational Linguistics: ACL 2025*.
6. Yuxia Wang, Artem Shelmanov, ..., **Tarek Mahmoud**, ..., and Preslav Nakov. “GenAI Content Detection Task 1: English and Multilingual Machine-Generated Text Detection: AI vs. Human”. In *Proceedings of the 1st Workshop on GenAI Content Detection: GenAIDetect 2025*.
7. Artur Muratov, Hana Fatima Shaikh, Vanshikaa Jani, **Tarek Mahmoud**, ..., and Preslav Nakov. “FRaN-X: FRaming and Narratives-eXplorer”. *arXiv preprint arXiv:2507.06974, 2025*.

8. **Tarek Mahmoud**, Zhuohan Xie, and Preslav Nakov. “BERTastic at SemEval-2025 Task 10: State-of-the-Art Accuracy in Coarse-Grained Entity Framing for Hindi News”. In *Proceedings of the 19th International Workshop on Semantic Evaluation: SemEval 2025*.
9. Yuxia Wang, Jonibek Mansurov, ..., **Tarek Mahmoud**, ..., and Preslav Nakov. “M4: Multi-Generator, Multi-Domain, and Multi-Lingual Black-Box Machine-Generated Text Detection”. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: EACL 2024. (Best Resource Paper Award)*
10. Yuxia Wang, Jonibek Mansurov, ..., **Tarek Mahmoud**, ..., and Preslav Nakov. “SemEval-2024 Task 8: Multidomain, Multimodel and Multilingual Machine-Generated Text Detection”. In *Proceedings of the 18th International Workshop on Semantic Evaluation: SemEval 2024*.
11. Yuxia Wang, Jonibek Mansurov, Petar Ivanov, ..., **Tarek Mahmoud**, ..., and Preslav Nakov. “M4GT-Bench: Evaluation Benchmark for Black-Box Machine-Generated Text Detection”. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics: ACL 2024*.
12. Mervat Abassy, Kareem Elozeiri, ..., **Tarek Mahmoud**, ..., and Preslav Nakov. “LLM-DetectAlve: A Tool for Fine-Grained Machine-Generated Text Detection”. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: EMNLP 2024. (Demo)*
13. Ahmed Sajwani, Ahmed El Setohy, Ahmed Mekky, ..., **Tarek Mahmoud**, ..., and Preslav Nakov. “FRAPPE: FRAMing, Persuasion, and Propaganda Explorer”. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: EACL 2024. (Demo)*
14. **Tarek Mahmoud**, and Preslav Nakov. “BERTastic at SemEval-2024 Task 4: State-of-the-Art Multilingual Propaganda Detection in Memes via Zero-Shot Learning with Vision-Language Models”. In *Proceedings of the 18th International Workshop on Semantic Evaluation: SemEval 2024*.
15. **Tarek Mahmoud**, and Preslav Nakov. “BERTastic at SemEval-2023 Task 3: Fine-Tuning Pretrained Multilingual Transformers – Does Order Matter?”. In *Proceedings of the 17th International Workshop on Semantic Evaluation: SemEval 2023*.

Patents

- 2017 **Water Tank Notifier**: US Patent 9,593,977 B2
- 2014 **Qibla Orientation Device**: US Patent 8,898,012 B2

Awards & Honors

- 2026 **ACL 2026 Oral Presentation & SAC Highlights Award**: “Uncovering Temporal Framing in the News” (Top 15%)
- 2024 **Best Resource Paper Award**: EACL 2024
- 2022–2023 **MBZUAI Executive Program**: nominated to facilitate AI and innovation workshops for HE-level government officials and C-level executives
- 2022–2026 **Merit-Based Full Fellowship**: MBZUAI
- 2022 **Smart Entrepreneur Award**: Eureka Innovation Lab, Presight
- 2019 **Best Air Team Award & ERL GSA Special Prize**: European Robotics League, Spain
- 2016–2019 **Merit-Based Full Fellowship**: KAUST
- 2016 **First Honors**: Electrical Engineering, GPA 4.00/4.00, KFUPM
- 2012–2016 **Merit-Based Full Scholarship**: KFUPM

Professional Service

- 2024–2025 **Workshop Organizer:** SemEval 2025 Task 10 (Multilingual Narrative Extraction), GenAIDetect 2025 Task 1 (AI vs. Human Text Detection), SemEval 2024 Task 8 (Machine-Generated Text Detection)
- 2022–present **Reviewer:** ACL, EMNLP, NAACL, EACL, SemEval